

Weekly Assignment: Predicates & Quantified Statements (Epp 3.1, 3.2, & 3.3)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 3.1 #5 b,d). Find the truth set of each predicate.

b. Predicate: $6/d$ is an integer, domain: $\mathbf{Z}^+$

d. Predicate: $1 \leq x^2 \leq 4$, domain: $\mathbf{Z}$

Problem 2 (Epp 3.1 #8). Let T be the set of all strings of length 3 consisting of 0's and 1's. List all the strings in T that satisfy the following two conditions simultaneously:

1. For every string s in T , the second character of s is 1 or the first two characters of s are the same.
2. No string in T has all three characters the same.

Problem 3 (Epp 3.1 #23 b). Rewrite the following statement in the two forms " $\forall$ _____ x , if _____ then _____" and " $\forall x$, _____" (without an if-then).

Every computer science student needs to take data structures.

Problem 4 (Epp 3.2 #3 b,d). Write a formal negation for each of the following statements.

b. $\forall$ computer c , c has a CPU.

d. $\exists$ a band b such that b has won at least 10 Grammy awards.

Problem 5 (Epp 3.2 #48). Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite the following without using the word *necessary* or *sufficient*:

Being a polynomial is not a sufficient condition for a function to have a real root.

Problem 6 (Epp 3.3 #16). Consider the following statement:

$\exists$ a real number u such that $\forall$ real number v , $uv = v$.

- (a) Rewrite this statement in English without using the symbol $\forall$ or $\exists$ or variables, expressing your answer as simply as possible.
- (b) Write a negation for the statement.

Problem 7 (Proof). Consider the following statement:

The difference of the squares of any two consecutive integers is odd.

- (a) Translate this statement into mathematical quantifiers.
- (b) Write the negation of your quantified statement from part (a).
- (c) Determine whether the original statement is true or false. If it is true, prove it; if it is false, prove its negation. Your proof may be a simple explanation in plain English, but it should be convincing and complete.

Hint: If n is any integer, then the two consecutive integers are n and $n + 1$. What is $(n + 1)^2 - n^2$?